

For a long-horizon investor, U.S. Treasuries are usually neither a risk-free asset nor the optimal liability-hedging asset. If the investment horizon exceeds the longest available maturity of the risk-free bond, which is the case for some sovereign wealth funds and family offices, then investors do not have access to a risk-free asset. Furthermore, many investors have liabilities denominated in *real*, not *nominal*, terms. But even long-horizon real bonds are not the optimal liability-hedging asset if there are other factors. For pension plans, these include longevity risk, economic growth, and credit risk. Individual investors may face inflation risks, like for medical care and college tuition, that are not adequately reflected in general CPI inflation. The liability hedging portfolio emphasizes what types of assets (or more broadly what kinds of factors, see chapter 14) pay off to meet the worst times of the investor, in terms of when and how the liabilities come due. If liabilities increase when credit spreads narrow, for example, as they do for pension funds, then the liability-hedging portfolio must hold large quantities of assets that are sensitive to credit risk.

What if you can't meet the liabilities in the first place? Sadly, this condition applies to many investors today, especially public pension funds. CalPERS, for example, only had a funding ratio (the ratio of assets to actuarial liabilities) of 65% at June 30, 2010. Strictly speaking, the Merton–Samuelson asset allocation advice outlined in sections 2 and 3 applies only after the liabilities can be met, both in terms of the present value of the liabilities and after the liability-hedging portfolio has been constructed. If assets are not sufficient to meet current liabilities, then the investor must face the fact that default will happen in some states of the world. Portfolios can be constructed to minimize this probability, but avoiding insolvency requires a different optimization than the maximization of utility examined in equation (4.2). In certain cases, it may be optimal for the investor to engage in risk-seeking behavior if the assets are far enough below the value of the liabilities. It is the Hail Mary pass; you have nothing to lose, and you are likely to go bankrupt anyway.<sup>35</sup>

## 4.2. POPULAR INVESTMENT ADVICE

The three types of portfolios for long-term investors

1. Liability-hedging portfolio;
2. Short-run, or myopic, market portfolio; and
3. Long-run opportunistic, or long-term hedging demand, portfolio

<sup>35</sup> Ang, Chen, and Sundaresan (2013) demonstrate that this behavior is optimal in a liability-driven investment context with downside risk. Andonov, Bauer, and Cremers (2012) show that U.S. public pension funds, which have discount rates based on the earning rates of their assets, have incentives to take more risk. They show that funds that are more underfunded have invested proportionately more in risky assets.